SPARQL

CQ.1 What are the modalities of APA (frequency, intensity) for physical activity and muscle building?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?prop ?modalite WHERE {
:RenforcementMusculaire rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) ?anonClass .
?anonClass owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ; owl:onProperty ?prop ; owl:hasValue ?modalite .

FILTER(?prop IN (:aPourFrequence, :aPourIntensite))
}
```

SPARQL query:

```
WHERE {
  :RenforcementMusculaire rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anonClass .
  ?anonClass owl:intersectionOf ?list .
  ?list rdf:rest*/rdf:first ?element .

  ?element a owl:Restriction ;
        owl:onProperty ?prop ;
        owl:hasValue ?modalite .

  FILTER(?prop IN (:aPourFrequence, :aPourIntensite))
}
```

prop	modalite
aPourFrequence	"par semaine"
aPourIntensite	Borg modéré

CQ2. What physical activities are prohibited in the context of inflammatory heart disease?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?activite WHERE {
:CardiopathiesInflammatoires rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list ] .
?list rdf:rest*/rdf:first ?element .
?element a owl:Restriction ;
owl:onProperty :aPourContreIndication ; owl:someValuesFrom ?activite .
}
```

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT DISTINCT ?activite
WHERE {
  :CardiopathiesInflammatoires rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list ] .
  ?list rdf:rest*/rdf:first ?element .
  ?element a owl:Restriction ;
        owl:onProperty :aPourContreIndication ;
        owl:someValuesFrom ?activite .
}
```

activite
Activité physique

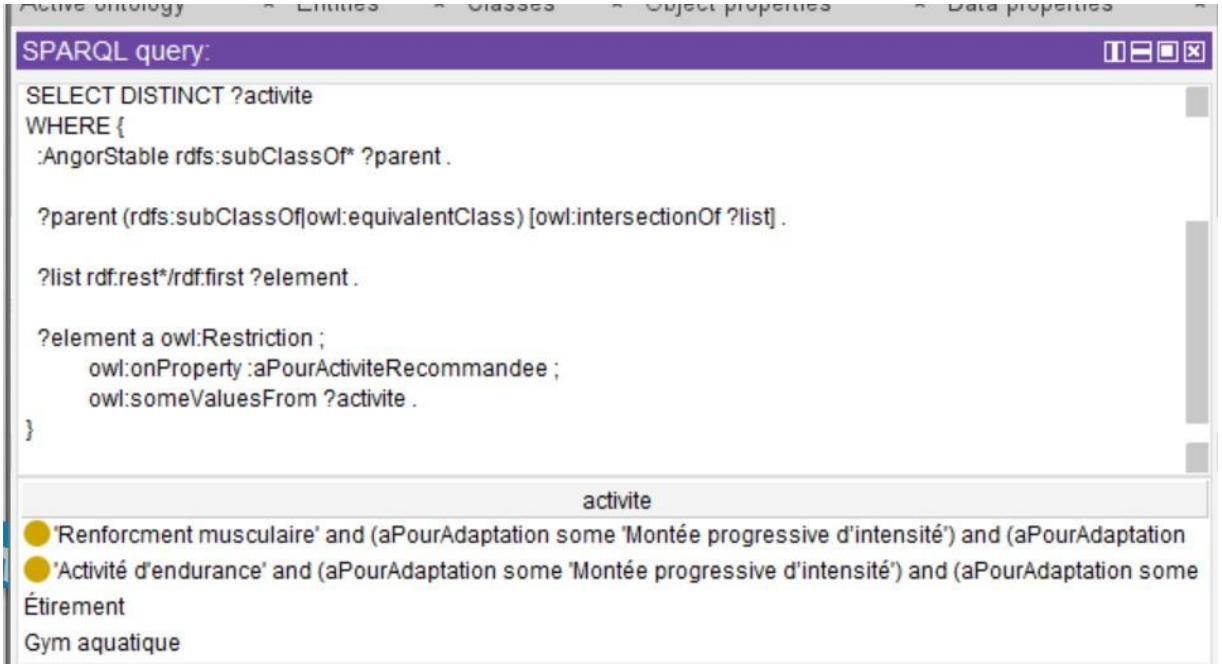
CQ3. What physical activities are recommended for stable angina pectoris?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?activite WHERE {
:AngorStable rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) [owl:intersectionOf ?list] .

?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ;
owl:onProperty :aPourActiviteRecommandee ; owl:someValuesFrom ?activite .
}
```



CQ4. What facilitator should be offered to a patient who does not have a gym near their home?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?levier WHERE {
:AbsenceGymnase rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?element .
?element a owl:Restriction ; owl:onProperty :aPourLevier ;
owl:someValuesFrom ?unionClass .
?unionClass owl:unionOf ?uList .
?uList rdf:rest*/rdf:first ?levier .
}
```

SPARQL query:

```
SELECT DISTINCT ?levier
WHERE {
  :AbsenceGymnase rdfs:subClassOf* ?parent .
  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .
  ?list rdf:rest*/rdf:first ?element .
  ?element a owl:Restriction ;
        owl:onProperty :aPourLevier ;
        owl:someValuesFrom ?unionClass .
  ?unionClass owl:unionOf ?uList .
  ?uList rdf:rest*/rdf:first ?levier .
}
```

levier
Activité à domicile
Activité sans matériel
Marche autour du domicile
Micro séance 10 min

CQ5. What contraindications to physical activity are associated with acute coronary syndrome in a patient also with hypertension and a stent?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>

SELECT DISTINCT ?profil ?contreIndication WHERE {
VALUES ?profil {
:SyndromeCoronarienAigu
:Hypertension_arterielle
}

?profil rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?restriction .

?restriction a owl:Restriction ;
owl:onProperty :aPourContreIndication ; owl:someValuesFrom ?contreIndication .
}
```

SPARQL query:

```
  :Hypertension_arterielle
 }

 ?profil rdfs:subClassOf* ?parent .
 ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
 ?anon owl:intersectionOf ?list .
 ?list rdf:rest*/rdf:first ?restriction .

 ?restriction a owl:Restriction ;
        owl:onProperty :aPourContreIndication ;
        owl:someValuesFrom ?contreIndication .
}
```

profil	contreIndication
Syndrome coronarien aigu	Activité physique
Hypertensiopon arterielle	Travail en isométrie
Hypertensiopon arterielle	Musculation lourde

CQ6. What equipment is needed to perform a leg abduction exercise?

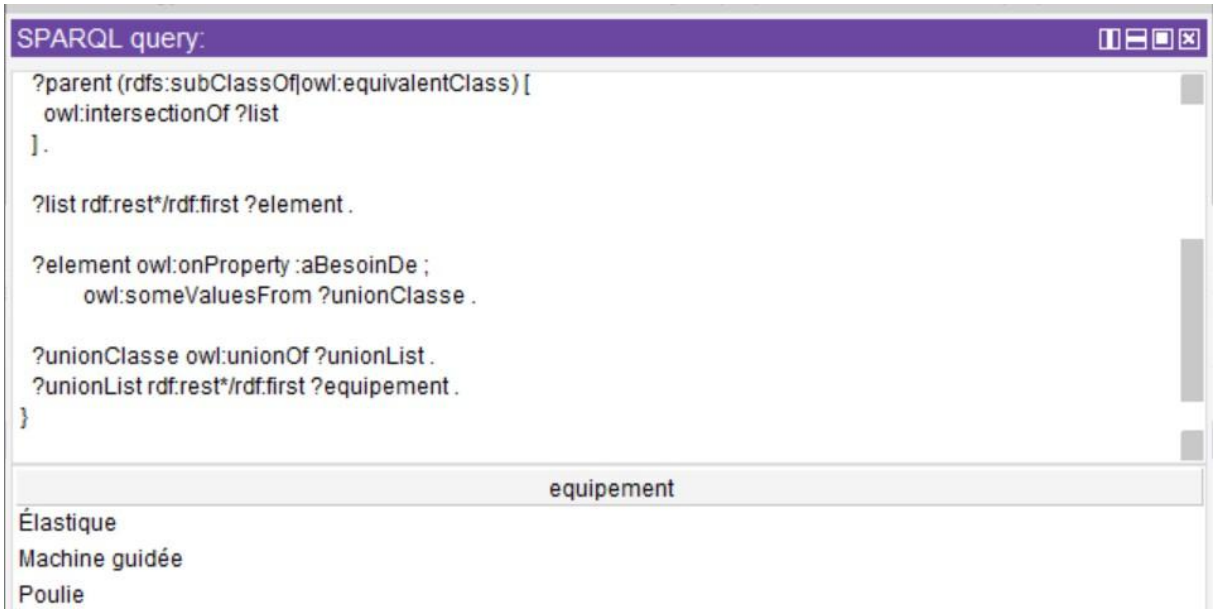
```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?equipement WHERE {
:AbductionDeJambe rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) [ owl:intersectionOf ?list
] .

?list rdf:rest*/rdf:first ?element .

?element owl:onProperty :aBesoinDe ; owl:someValuesFrom ?unionClasse .

?unionClasse owl:unionOf ?unionList .
?unionList rdf:rest*/rdf:first ?equipement .
}
```



CQ7. Which physical activities require adaptation in the context of functional coronary artery disease?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?actFiller ?adaptation WHERE {
:CoronaropathieFonctionnelle rdfs:subClassOf* ?parent .
?parent (rdfs:subClassOf|owl:equivalentClass) ?ce .
?ce owl:intersectionOf ?list .
?list rdf:rest*/rdf:first ?elemAct .

?elemAct a owl:Restriction ;
owl:onProperty :aPourActiviteRecommandee ; owl:someValuesFrom ?actFiller .

?actFiller owl:intersectionOf ?actList2 .
?actList2 rdf:rest*/rdf:first ?adRest .

?adRest a owl:Restriction ;
owl:onProperty :aPourAdaptation .
{
?adRest owl:someValuesFrom ?adaptation .
FILTER NOT EXISTS { ?adaptation owl:unionOf ?u }
} UNION
{
?adRest owl:someValuesFrom ?uClass .
?uClass owl:unionOf ?u .
?u rdf:rest*/rdf:first ?adaptation .
}
}
ORDER BY ?actFiller ?adaptation
```

SPARQL query:

```
 {
  ?adRest owl:someValuesFrom ?adaptation .
  FILTER NOT EXISTS { ?adaptation owl:unionOf ?u }
 }
 UNION
 {
  ?adRest owl:someValuesFrom ?uClass .
  ?uClass owl:unionOf ?u .
  ?u rdf:rest*/rdf:first ?adaptation .
 }
}
ORDER BY ?actFiller ?adaptation
```

actFiller	adaptation
'Renforcment musculaire' and (aPourAdaptation some	Montée progressive d'intensité
'Renforcment musculaire' and (aPourAdaptation some	Réduction de charge
'Activité d'endurance' and (aPourAdaptation some 'Mor	Borg faible
'Activité d'endurance' and (aPourAdaptation some 'Mor	Borg modéré
'Activité d'endurance' and (aPourAdaptation some 'Mor	Montée progressive d'intensité

CQ8. What are the objectives associated with stretching exercises?

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT DISTINCT ?objectif WHERE {
:Etirement rdfs:subClassOf* ?parent .

?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
?anon owl:intersectionOf ?list .

?list rdf:rest*/rdf:first ?element .

?element a owl:Restriction ; owl:onProperty :aPourObjectif ;
owl:someValuesFrom ?objectif .
}
```

SPARQL query:

```
WHERE {
  :Etirement rdfs:subClassOf* ?parent .

  ?parent (rdfs:subClassOf|owl:equivalentClass) ?anon .
  ?anon owl:intersectionOf ?list .

  ?list rdf:rest*/rdf:first ?element .

  ?element a owl:Restriction ;
        owl:onProperty :aPourObjectif ;
        owl:someValuesFrom ?objectif .
}
```

objectif
Amélioration de l'amplitude

CQ9. What physical exercises were prescribed for X?

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/APA4CAD#>
SELECT ?prescription ?date ?activite
WHERE {
?prescription :concerne :Patient1 .
?prescription :aPourDate ?date .
?prescription :contient ?activite .
}
```

SPARQL query:

```
PREFIX : <http://www.semanticweb.org/mmolina/ontologies/2025/11/untitled-ontology-50#>

SELECT ?prescription ?date ?activite
WHERE {
  ?prescription :concerne :Patient1 .
  ?prescription :aPourDate ?date .
  ?prescription :contient ?activite .
}
```

prescription	date	activite
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	ActivitéEndurancePatient1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Renforcement musculaire Patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Etirement patient 1
Prescription1	"2026-02-09T14:30:00"^^<http://www.v	Sport_de_combat_Patient_1